

22: Binomial Logistic Regression

Stat 230 - Spring 2026

1 Data overview

Orobanche is a genus of parasitic plants without chlorophyll that grow on the roots of flowering plants. In an experiment, seeds from two varieties, *Orobanche aegyptiaca* 75 and *Orobanche aegyptiaca* 73, were brushed with extract from either a cucumber root or a bean root. Researchers recorded the number of seeds that eventually germinated.

Run the code chunk below to load the data:

```
orobanche <- read.csv("https://aluby.github.io/stat230/data/orobanche.csv")
```

The variables in the data set are:

- *y*: count of seeds germinated
- *n*: number of seeds
- *variety*: either oa75 or oa73
- *root*: either cucumber or bean

1. Explain why this is an example of a binomial response as compared to a binary response
2. Does there appear to be a relationship between the proportion of seeds that germinate and variety? What about root?
 - i. create boxplots of the estimated proportion by each variable.
 - ii. create boxplots of the estimated proportion using one variable on the x-axis and the other as the fill color.

What do you learn from these plots?

2 Additive model

To begin, let's fit an additive logistic regression model where the main effects for `variety` and `root` are both included. Like in binary logistic regression, we use the `glm()` function to fit the model and specify `family = binomial`. The difference is in how we specify the response variable. For binomial logistics regression we specify the response as a proportion (e.g., y/n), and we need to use the `weights` argument to indicate the number of trials (i.e., n).

3 Fill in the blanks in the below code chunk to fit the additive binomial logistic regression model:

```
oro_glm1 <- glm(y/n ~ variety + root, weights = n, family = binomial, data = orobanch
```

3. Report the fitted equation for the linear predictor (i.e., the log-odds of germination).

4. Interpret the coefficient for `varietyoa75` in context.

5. We can create plots of the Pearson or deviance residuals using the `resid_panel` function from the `car` package and setting `type = "pearson"` or `type = "deviance"`. Create deviance residual plots for the fitted model. Do you see any model deficiencies?

```
ggResidpanel::resid_panel(oro_glm1, type = "deviance", plots = c("resid", "qq"), smoo
```

3 Adding an interaction term

You should have found evidence of lack of fit, so we may be missing a term. The only other thing we can add here is an interaction.

6. Fit a new model that includes an interaction

```
oro_glm2 <- glm(____ ~ variety*root, weights = ____, family = binomial, data = orobanch
```

7. Create deviance residual plots for the fitted model. Do you see any model deficiencies?

```
# Put your residual plot code here
```

8. Perform a Wald-based test to determine if the interaction is needed.

9. Perform a Likelihood Ratio Test to compare the two models.

10. Do the results from 8 and 9 agree? What do you conclude?

4 Function quick reference

The following table summarizes the functions discussed above:

Function	Purpose
<code>glm()</code>	Fit generalized linear models, including binomial logistic regression models.
<code>resid_panel()</code>	Create residual plots to assess model fit. Set either <code>type = "pearson"</code> or <code>type = "deviance"</code> for binomial regression models.